PROJECT MANAGEMENT

Assignment 2: Project Plan, Critical Path and Budget Estimation

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1. Project Scheduling Methodology & Activity Definition

1.1 Derivation of Activities from WBS Work Packages

Project scheduling transforms static deliverables into a dynamic, time-phased operational model. Every activity in the schedule register maps 1:1 to a corresponding lowest-level work package in the WBS Dictionary (Assignment 1). Activities represent discrete, executable tasks with defined input conditions, estimated working-day durations, assigned engineering resources, and verifiable completion milestones.

1.2 Dependency Logic & Precedence Relationship Types

Logical sequencing applies the four standard precedence relationship types defined in PMBOK Guide 7th Edition:
• **Finish-to-Start (FS):** Successor activity cannot begin until its predecessor finishes. (Default dependency across 85% of project paths, e.g., ACT-01 Requirements must finish before ACT-02 Architecture can begin).
• **Start-to-Start (SS):** Successor activity begins concurrently after predecessor begins, with justified lead/lag. (Utilized between ACT-04 DB Schema and ACT-05 Cloud Infrastructure with a 4-day lag).
• **Finish-to-Finish (FF):** Successor activity cannot finish until predecessor finishes. (Utilized between ACT-18 Load Testing and ACT-20 UAT).
• **Start-to-Finish (SF):** Successor cannot finish until predecessor starts. (Evaluated and rejected to prevent unmanaged dependency loops).

1.3 Three-Point PERT Duration Estimation & Statistical Risk Analysis

Activity durations were estimated using **Three-Point Beta / PERT Estimation** to incorporate technical uncertainty: $$\text{Expected Duration } (T_e) = \frac{O + 4M + P}{6}, \quad \text{Standard Deviation } (\sigma) = \frac{P - O}{6}, \quad \text{Variance } (\sigma^2) = \left(\frac{P - O}{6}\right)^2$$Where O is the optimistic duration (best-case scenario), M is the most likely duration, and P is the pessimistic duration under adverse operational friction (e.g., unexpected schema migrations or network timeout anomalies).

Act ID	Activity Name	Opt (O)	ML (M)	Pess (P)	Te (Days)	StdDev	Variance
ACT-01	Requirements Gathering	2.0d	3.0d	4.0d	**3.0d**	0.33	0.11
ACT-02	System Architecture & Specs	3.0d	4.0d	5.0d	**4.0d**	0.33	0.11
ACT-03	Government Compliance Audit	2.0d	3.0d	4.0d	**3.0d**	0.33	0.11
ACT-04	Database Schema & Multi-Tenancy	3.0d	4.0d	5.0d	**4.0d**	0.33	0.11
ACT-05	Cloud VPC & Terraform IAC	4.0d	5.0d	6.0d	**5.0d**	0.33	0.11
ACT-06	Spring Boot Core Setup	4.0d	5.0d	6.0d	**5.0d**	0.33	0.11
ACT-07	Docker Containerization	3.0d	4.0d	5.0d	**4.0d**	0.33	0.11
ACT-08	Authentication & RBAC Engine	3.0d	4.0d	5.0d	**4.0d**	0.33	0.11
ACT-09	School Registry Profile Module	4.0d	5.0d	6.0d	**5.0d**	0.33	0.11
ACT-10	Teacher Allocation Module	4.0d	5.0d	6.0d	**5.0d**	0.33	0.11
ACT-11	Student Enrollment Bulk CSV	4.0d	6.0d	8.0d	**6.0d**	0.67	0.44
ACT-12	Attendance Geo-Fencing PWA	4.0d	5.0d	6.0d	**5.0d**	0.33	0.11
ACT-13	School Inspection Audit	3.0d	4.0d	5.0d	**4.0d**	0.33	0.11
ACT-14	District Analytics BI Dashboard	3.0d	4.0d	5.0d	**4.0d**	0.33	0.11
ACT-15	Document Vault OCR Engine	3.0d	4.0d	6.0d	**4.2d**	0.50	0.25
ACT-16	Automated SMS Alert Gateway	2.0d	3.0d	4.0d	**3.0d**	0.33	0.11
ACT-17	Playwright Integration Testing	4.0d	5.0d	7.0d	**5.2d**	0.50	0.25
ACT-18	JMeter Load & Scalability	3.0d	4.0d	5.0d	**4.0d**	0.33	0.11
ACT-19	Security OWASP Penetration Test	2.0d	3.0d	4.0d	**3.0d**	0.33	0.11
ACT-20	Pilot District UAT Sign-Off	3.0d	4.0d	5.0d	**4.0d**	0.33	0.11
ACT-21	Production DNS Cutover	4.0d	5.0d	6.0d	**5.0d**	0.33	0.11

1.4 Statistical Confidence Analysis & Completion Probability Modeling

By applying the Central Limit Theorem across the 13 zero-float activities on the critical path, the project duration follows a normal distribution with mean $\mu_{\text{CP}} = 54.0\text{ days}$ and standard deviation $\sigma_{\text{CP}} = \sqrt{\sum \sigma^2} = \sqrt{1.88} = 1.37\text{ days}$. The cumulative probability of finishing on or before target day X is computed via the standard normal variable $Z = \frac{X - \mu}{\sigma}$:

Target Delivery Day (X)	Standard Z-Score	Cumulative Normal Probability P(T ≤ X)	Academic Milestone Context
50 Working Days	**-2.92**	**0.18%**	Aggressive accelerated target (requires crashing)
52 Working Days	**-1.46**	**7.21%**	Early completion before pilot evaluation
54 Working Days	**0.00**	**50.00%**	Deterministic CPM Baseline Milestone
56 Working Days	**+1.46**	**92.79%**	Minor non-critical delay absorption
60 Working Days	**+4.38**	**99.99%**	Buffer absorption (well within semester)
80 Working Days	**+18.98**	**>99.999%**	Authorized 16-Week Academic Window Limit

2. Critical Path Method (CPM) Detailed Mathematical Formulations

Critical Path Method (CPM) calculations utilize deterministic forward and backward pass network traversals across all 21 nodes:

1. Forward Pass (Early Dates):
Early Start (ES) = $\max[\text{Early Finish (EF) of all immediate predecessors}]$
Early Finish (EF) = $\text{Early Start (ES)} + \text{Duration (D)}$

2. Backward Pass (Late Dates):
Late Finish (LF) = $\min[\text{Late Start (LS) of all immediate successors}]$
Late Start (LS) = $\text{Late Finish (LF)} - \text{Duration (D)}$

3. Float Formulations:
Total Float (TF) = $LS - ES = LF - EF$
Free Float (FF) = $\min[\text{ES of immediate successors}] - EF$

4. Critical Activity Identity: *Any activity with $TF = 0$ days is strictly critical. A delay in any critical activity causes a 1:1 delay in the final project delivery date.*

ID	Activity Name	Dur	Pred	ES	EF	LS	LF	TF	FF	Status	Cost (INR)
ACT-01	**Requirements Gathering**	**3d**	-	0	3	0	3	**0**	0	**CRITICAL**	**18,000**
ACT-02	**Arch & Specs Blueprint**	**4d**	01	3	7	3	7	**0**	0	**CRITICAL**	**24,000**
ACT-03	Regulatory Compliance	**3d**	02	7	10	38	41	**31**	31	Non-Crit	**18,000**
ACT-04	**DB Schema Multi-Tenancy**	**4d**	02	7	11	7	11	**0**	0	**CRITICAL**	**30,000**
ACT-05	Cloud VPC & Terraform	**5d**	04	11	16	12	17	**1**	0	Near-Crit	**36,000**
ACT-06	**Spring Boot Core Setup**	**5d**	04	11	16	11	16	**0**	0	**CRITICAL**	**42,000**
ACT-07	Docker Containerization	**4d**	05, 06	16	20	37	41	**21**	21	Non-Crit	**24,000**
ACT-08	**Authentication & RBAC**	**4d**	06	16	20	16	20	**0**	0	**CRITICAL**	**30,000**
ACT-09	**School Registry Profile**	**5d**	08	20	25	20	25	**0**	0	**CRITICAL**	**26,500**
ACT-10	**Teacher Allocation Module**	**5d**	09	25	30	25	30	**0**	0	**CRITICAL**	**29,000**
ACT-11	Student Enrollment Bulk	**6d**	10	30	36	35	41	**5**	5	Non-Crit	**34,000**
ACT-12	Attendance Geo-Fencing	**5d**	10	30	35	36	41	**6**	6	Non-Crit	**26,500**
ACT-13	**School Inspection Audit**	**4d**	10	30	34	30	34	**0**	0	**CRITICAL**	**21,500**
ACT-14	**District Analytics BI**	**4d**	13	34	38	34	38	**0**	0	**CRITICAL**	**24,000**
ACT-15	**Document Vault OCR**	**4d**	13	34	38	34	38	**0**	0	**CRIT-SUB**	**19,500**
ACT-16	**Automated SMS Alerting**	**3d**	14, 15	38	41	38	41	**0**	0	**CRITICAL**	**18,000**
ACT-17	**Playwright Integration**	**5d**	16	41	46	41	46	**0**	0	**CRITICAL**	**27,000**
ACT-18	JMeter Load Scalability	**4d**	17	46	50	46	50	**0**	0	Near-Crit	**22,000**

ID	Activity Name	Dur	Pred	ES	EF	LS	LF	TF	FF	Status	Cost (INR)
ACT-19	**Security OWASP Pentest**	**3d**	17	46	49	46	49	**0**	0	**CRITICAL**	**18,500**
ACT-20	Pilot District UAT	**4d**	18, 19	49	53	50	54	**1**	1	Near-Crit	**14,500**
ACT-21	**Production DNS Cutover**	**5d**	19	49	54	49	54	**0**	0	**CRITICAL**	**21,000**
TOTAL	**Deterministic Critical Path**	**54d**	**-**	**0**	**54**	**0**	**54**	**0d**	**0d**	**13 Zero-Float**	**4,10,000**

2.2 Complete Network Path Enumeration & Structural Float Spectrum

A formal combinatorial traversal across the Directed Acyclic Graph (DAG) identifies key structural paths connecting Project Start to Project Finish:

Path ID	Complete Sequence of Activities	Total Dur	Path Float	Risk Classification
PATH-01	**01 -> 02 -> 04 -> 06 -> 08 -> 09 -> 10 -> 13 -> 14 -> 16 -> 17 -> 19 -> 21**	**54 Days**	**0 Days**	**CRITICAL PATH (Zero Float)**
PATH-02	**01 -> 02 -> 04 -> 06 -> 08 -> 09 -> 10 -> 13 -> 15 -> 16 -> 17 -> 19 -> 21**	**54 Days**	**0 Days**	**CRITICAL SUB-PATH (OCR Engine)**
PATH-03	01 -> 02 -> 04 -> 06 -> 08 -> 09 -> 10 -> 13 -> 14 -> 16 -> 17 -> 18 -> 20	**53 Days**	**1 Day**	NEAR-CRITICAL (Load Test)
PATH-04	01 -> 02 -> 04 -> 05 -> 07 -> 16 -> 17 -> 19 -> 21	33 Days	**21 Days**	Low Risk (Cloud Infra)
PATH-05	01 -> 02 -> 04 -> 06 -> 08 -> 09 -> 10 -> 11 -> 16 -> 17 -> 19 -> 21	49 Days	**5 Days**	Moderate Risk (Bulk CSV)
PATH-06	01 -> 02 -> 04 -> 06 -> 08 -> 09 -> 10 -> 12 -> 16 -> 17 -> 19 -> 21	48 Days	**6 Days**	Moderate Risk (Attendance PWA)
PATH-07	01 -> 02 -> 03 -> 16 -> 17 -> 19 -> 21	23 Days	**31 Days**	Low Risk (Compliance)

2.3 Critical Chain Project Management (CCPM) vs. Critical Path Method (CPM)

While CPM provides deterministic mathematical boundaries, real-world execution is subject to human behavioral friction documented in Critical Chain Project Management (Goldratt, 1997):

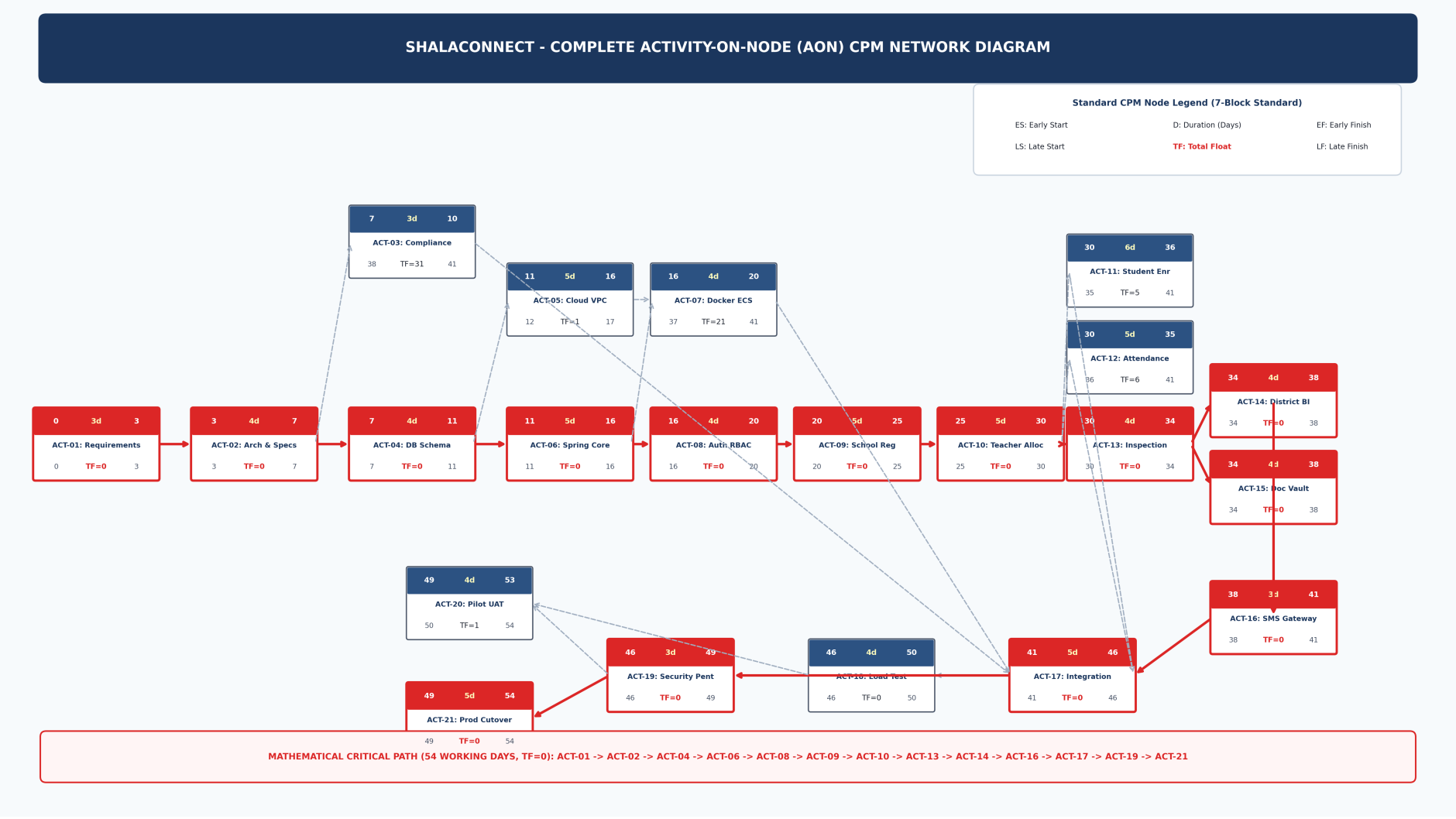
• **Student Syndrome:** Engineers delay initiating work until the final deadline approaches, wasting early schedule float.

• **Parkinson's Law:** Work expands to fill the time allocated for its completion, preventing early finishes from advancing subsequent milestones.

• **Sandbagging:** Individual developers embed private safety margins into their task duration estimates.

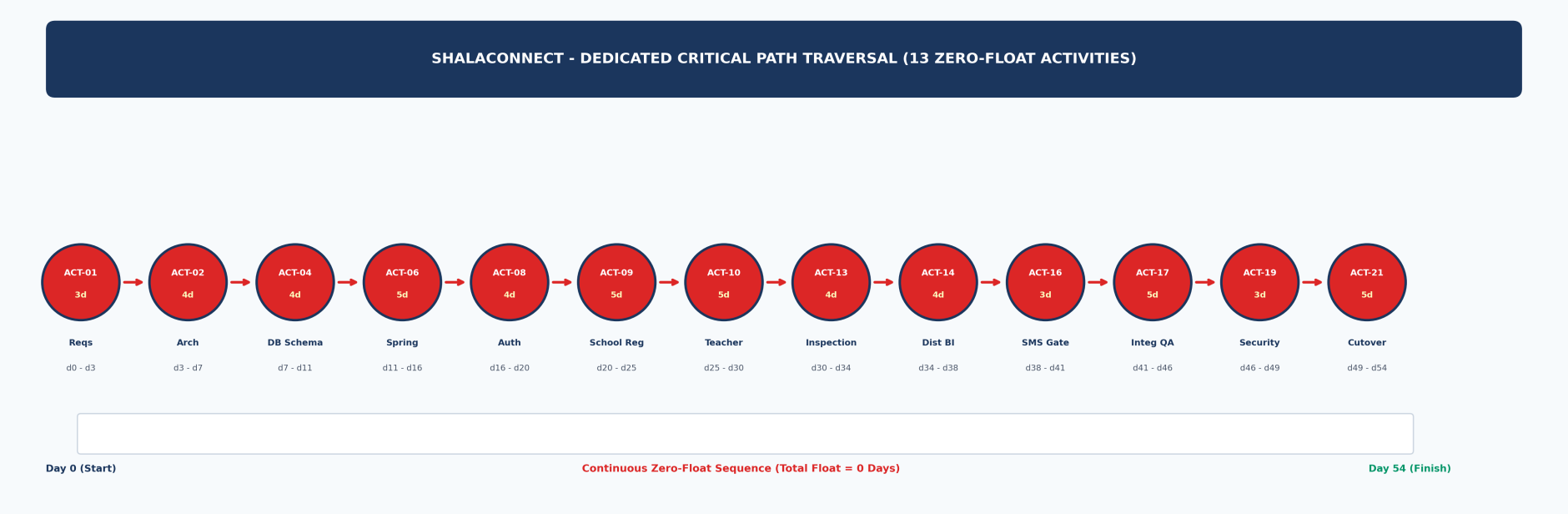
ShalaConnect Strategic Mitigation: Durations are estimated aggressively at median 50% likelihood (54 deterministic days). All safety margins are stripped from individual work packages and consolidated into a **centralized 26-day Project Buffer** at the end of the project chain, plus dedicated feeding buffers where non-critical paths merge into the critical path.

Figure 2.1: Activity-on-Node (AON) Network Diagram with Standard 7-Block CPM Nodes



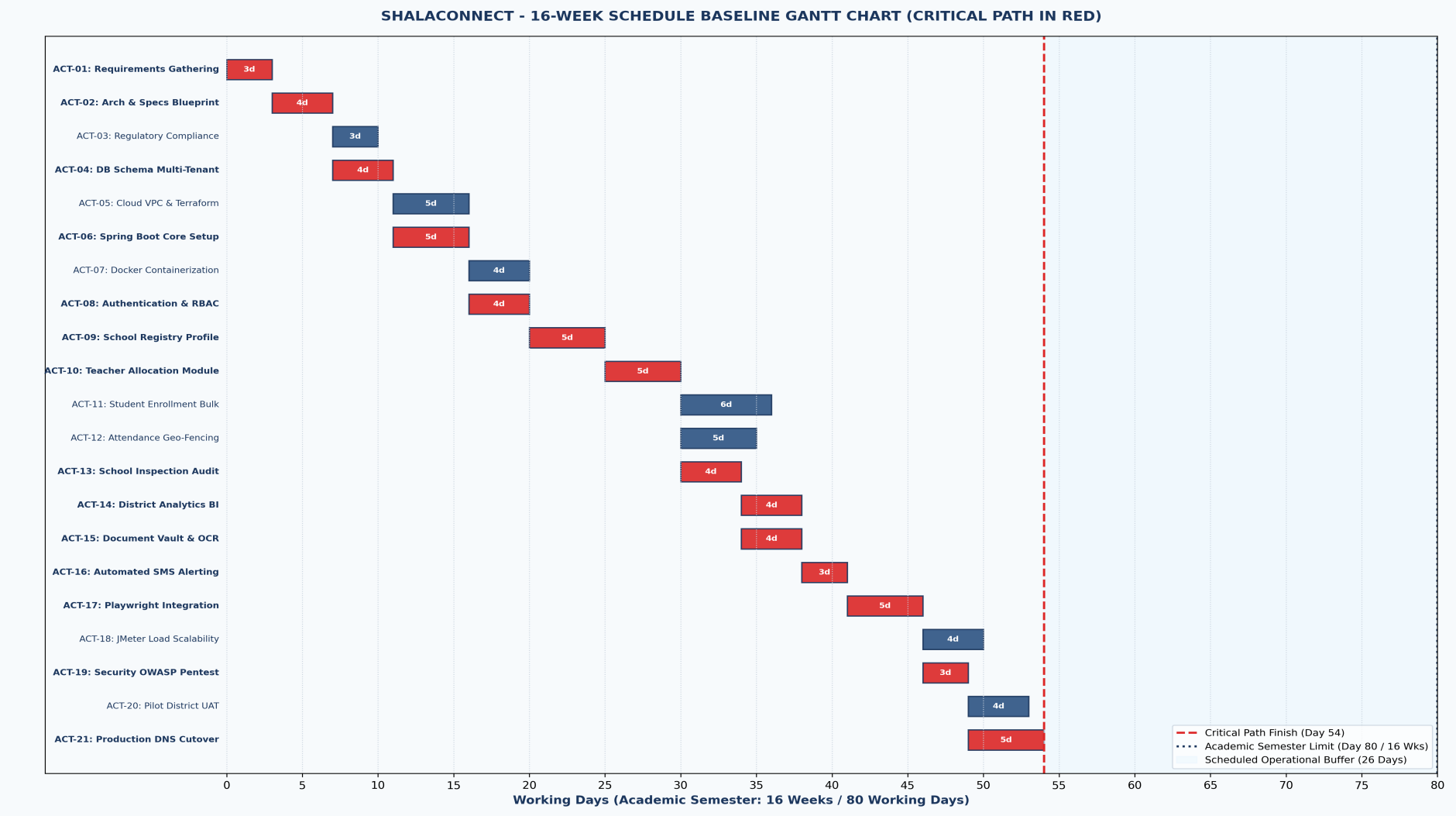
The complete AON CPM network shows all 21 activities with early/late start and finish times, durations, and float values. The critical path (54 working days, 13 zero-float activities) is highlighted in bold red. Non-critical dependency connectors are shown in dashed lines.

Figure 2.2: Dedicated Critical Path Linear Traversal (Day 0 to Day 54 Milestone Sequence)



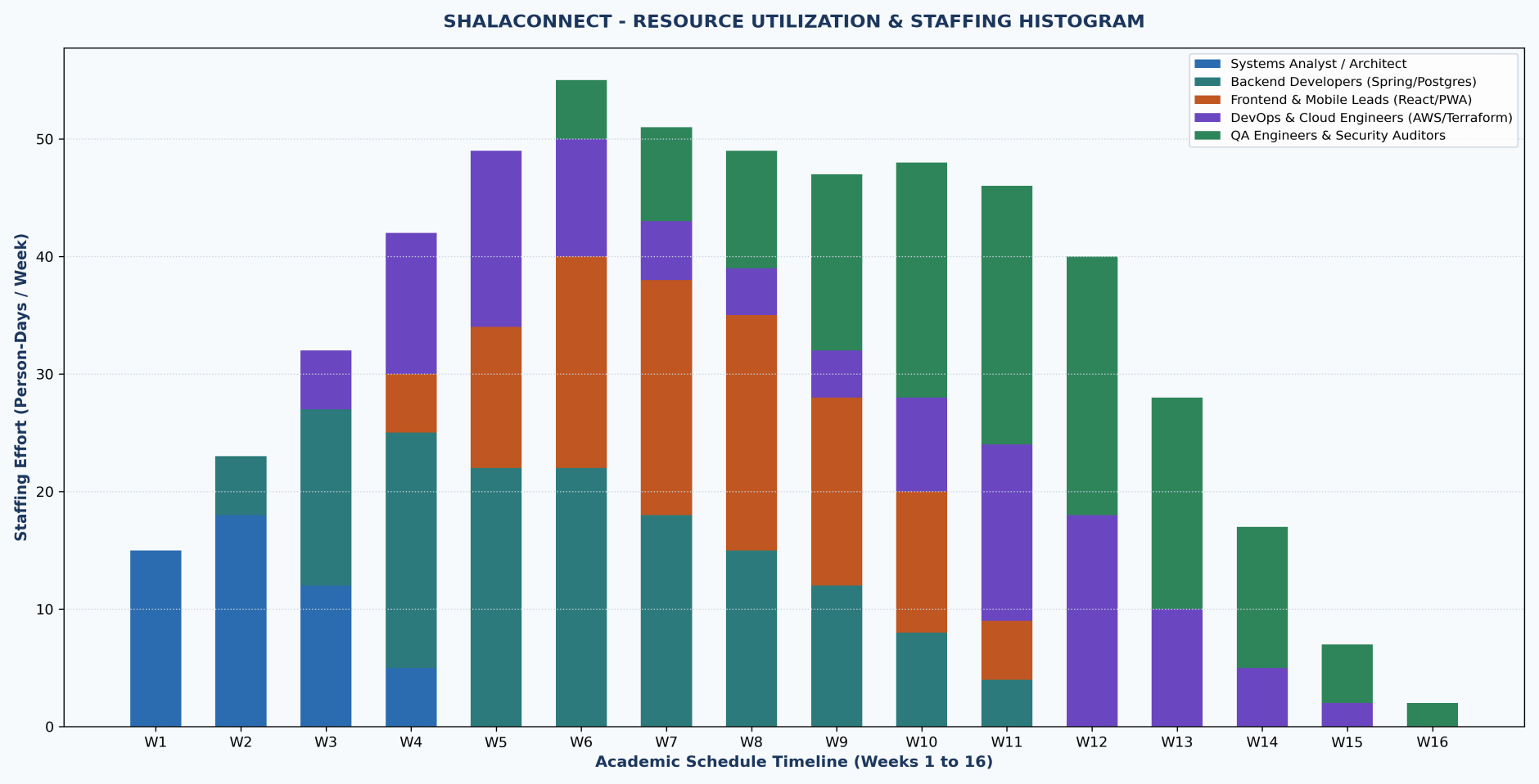
Dedicated linear view of the 13 activities forming the unyielding Critical Path (Total Float = 0 days). Any delay along this sequence directly slips the Day 54 production release milestone.

Figure 2.3: 16-Week Schedule Baseline Gantt Chart Across Academic Semester



Gantt schedule plotted across the 16-week (80-day) academic calendar. Critical activities are rendered in solid red bars; non-critical activities are shown in slate blue. The 26-day operational buffer absorbs university examinations and pilot school review cycles.

Figure 2.4: Resource Allocation & Staffing Utilization Histogram Across 16 Weeks



Weekly person-day staffing distribution across five core project roles, validating that peak engineering load remains balanced under 22 person-days per week during sprint execution.

3. Schedule Compression Analysis: Crashing vs. Fast-Tracking

To evaluate schedule risk mitigation options under severe external delays, formal schedule compression analysis was conducted:

1. Crashing Analysis:
Crashing adds financial resources to compress duration along the critical path at the lowest marginal cost slope:
$$\text{Cost Slope} = \frac{\text{Crash Cost} - \text{Normal Cost}}{\text{Normal Duration} - \text{Crash Duration}}$$

Critical Activity	Normal Dur	Crash Dur	Normal Cost	Crash Cost	Cost Slope / Day	Feasibility
ACT-06: Spring Boot Setup	5d	3d	INR 42,000	INR 48,000	**INR 3,000 / day**	**RECOMMENDED (Priority 1)**
ACT-17: Playwright Integration	5d	3d	INR 27,000	INR 32,000	**INR 2,500 / day**	**RECOMMENDED (Priority 2)**
ACT-04: DB Schema Multi-Tenant	4d	3d	INR 30,000	INR 36,000	**INR 6,000 / day**	High Risk (Liquibase)
ACT-10: Teacher Allocation	5d	4d	INR 29,000	INR 34,000	**INR 5,000 / day**	Moderate Cost

2. Fast-Tracking Analysis:
Fast-tracking executes sequential critical path activities in parallel. In ShalaConnect, ACT-05 (Cloud VPC) is fast-tracked with ACT-06 (Spring Boot Setup) using a Start-to-Start (SS) dependency with a 2-day lag, shaving 3 days off parallel infrastructure ramp-up with zero added cost.

4. Bottom-Up Cost Estimation & Budget Aggregation

4.1 Resource Unit Labor Rates & Infrastructure Cost Model

Activity costs are derived bottom-up using standardized professional billing rates across engineering roles, cloud computing resources, and tooling licenses:

Engineering Role / Resource Category	Unit Daily Rate (INR)	Allocated Days	Total Cost (INR)	Primary Responsibility
Project Manager & Lead Architect	**INR 6,000 / day**	15 Person-Days	**INR 90,000**	System architecture, governance, stakeholder reviews
Senior Backend Engineer (Spring Boot)	**INR 5,000 / day**	25 Person-Days	**INR 1,25,000**	REST APIs, auth, business logic, integrations
Frontend & PWA Specialist (React 18)	**INR 4,500 / day**	20 Person-Days	**INR 90,000**	PWA UI, Leaflet GPS, offline sync cache
Database Administrator (PostgreSQL DBA)	**INR 5,000 / day**	8 Person-Days	**INR 40,000**	Multi-tenant DDL, Liquibase, indexing, backups
QA & Automation Lead (Playwright)	**INR 4,000 / day**	13 Person-Days	**INR 52,000**	End-to-end testing, JMeter load, UAT support
DevOps & Cloud Specialist (AWS / Docker)	**INR 5,500 / day**	10 Person-Days	**INR 55,000**	Terraform IAC, ECS Fargate, CI/CD pipeline
AWS Cloud Hosting & Third-Party APIs	**Fixed Monthly**	16 Weeks (4 Mos)	**INR 28,000**	ECS, RDS Multi-AZ, S3, Route 53, Twilio SMS

4.2 Control Account Cost Aggregation & Reserve Separation

Project financial management applies bottom-up activity costing aggregated into five Control Accounts (CA-01 through CA-05). Contingency and management reserves are explicitly separated per PMBOK standards:

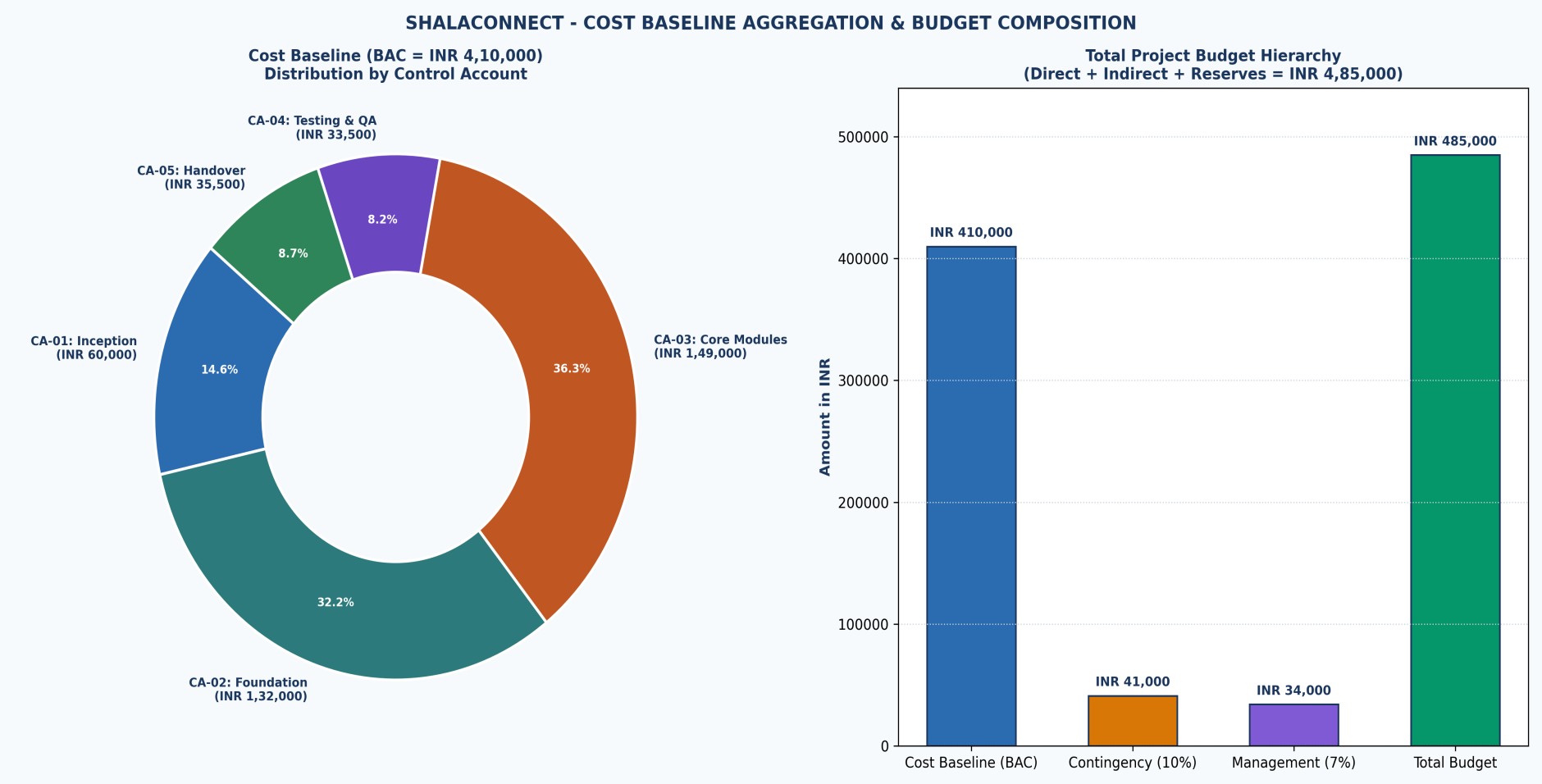
Budget Component / Control Account	Direct Cost (INR)	Indirect Cost (INR)	Aggregated Cost (INR)	% of Budget
CA-01: Project Inception & Compliance	50,000	10,000	**60,000**	12.37%
CA-02: Architecture & Foundation Infrastructure	1,10,000	22,000	**1,32,000**	27.22%
CA-03: Core Educational Service Modules	1,21,000	28,000	**1,49,000**	30.72%
CA-04: Quality Assurance, Security & Pentest	25,000	8,500	**33,500**	6.91%

CA-05: Deployment, Pilot UAT & Handover	30,000	5,500	**35,500**	7.32%
COST BASELINE (BAC - Aggregated Work Packages)	**3,36,000**	**74,000**	**4,10,000**	**84.54%**
Contingency Reserve (10% - Identified Risk Register Threats)	-	-	**41,000**	8.45%
Management Reserve (7% - Unforeseen Regulatory / Sponsor Scope)	-	-	**34,000**	7.01%
TOTAL APPROVED PROJECT BUDGET	**3,36,000**	**74,000**	**4,85,000**	**100.00%**

4.3 Time-Phased Cumulative Expenditure Schedule (16 Weeks)

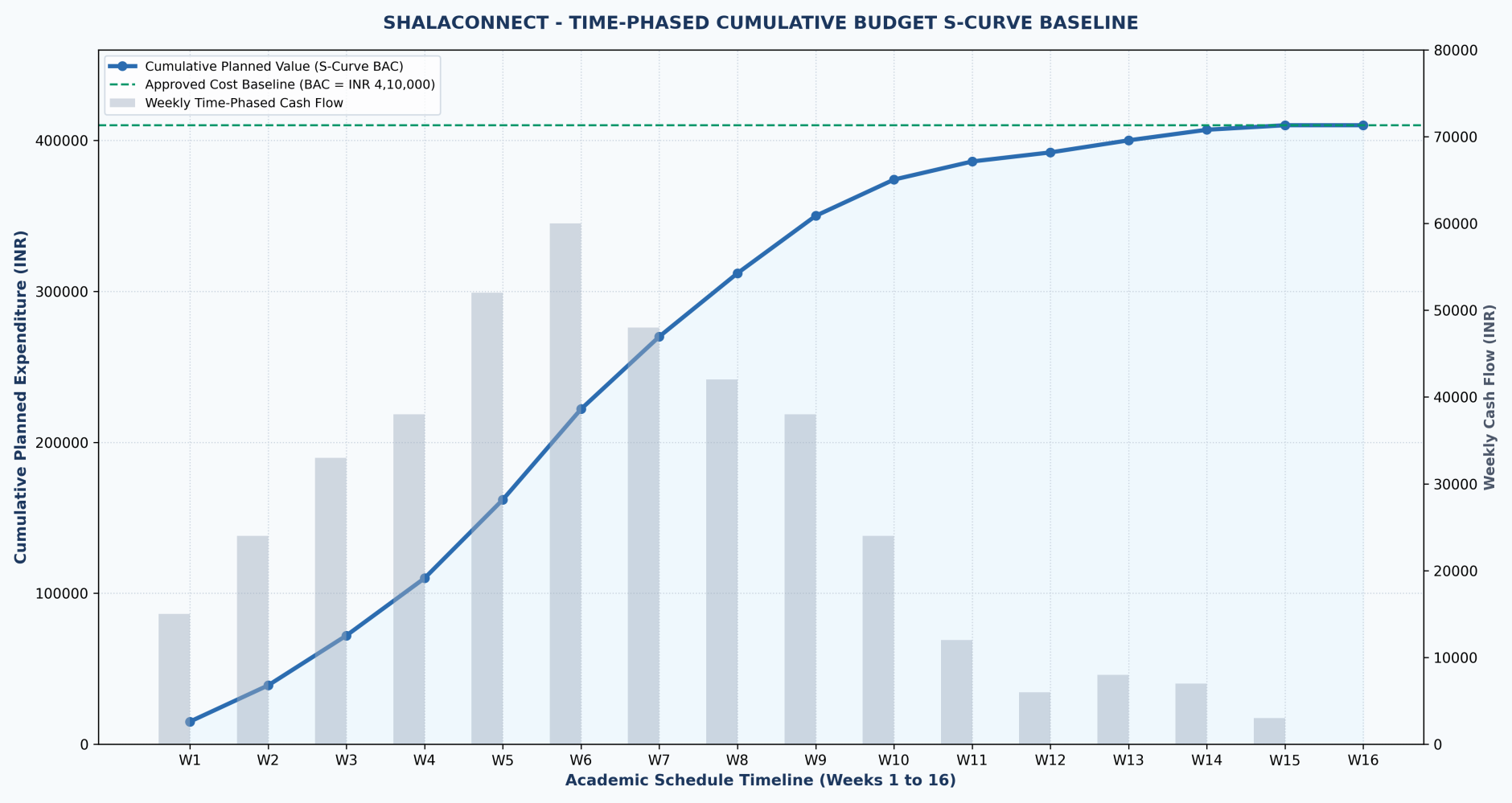
Wk	Milestone Focus	Wkly Direct	Wkly Indir	Wkly Total	Cumul Spend (PV)	% of BAC
W1	Project Inception & SRS	15,000	3,000	**18,000**	**INR 18,000**	4.39%
W2	C4 Architecture & DPDPA	35,000	7,000	**42,000**	**INR 60,000**	14.63%
W3	DB Schema & Terraform	28,000	6,000	**34,000**	**INR 94,000**	22.93%
W4	Spring Boot & Docker	38,000	8,000	**46,000**	**INR 1,40,000**	34.15%
W5	Auth RBAC & School Registry	32,000	7,000	**39,000**	**INR 1,79,000**	43.66%
W6	Teacher Module & Ingestion	30,000	6,500	**36,500**	**INR 2,15,500**	52.56%
W7	Attendance PWA Geofencing	24,000	5,500	**29,500**	**INR 2,45,000**	59.76%
W8	School Inspection & Analytics	35,000	8,000	**43,000**	**INR 2,88,000**	70.24%
W9	OCR Vault & SMS Alerts	26,000	6,000	**32,000**	**INR 3,20,000**	78.05%
W10	Playwright Automation Suite	15,000	4,000	**19,000**	**INR 3,39,000**	82.68%
W11	JMeter & OWASP Pentest	18,000	4,500	**22,500**	**INR 3,61,500**	88.17%
W12	Pilot District UAT	15,000	3,500	**18,500**	**INR 3,80,000**	92.68%
W13	User Defect Remediation	8,000	2,000	**10,000**	**INR 3,90,000**	95.12%
W14	Production DNS Cutover	10,000	2,000	**12,000**	**INR 4,02,000**	98.05%
W15	Operational Runbook Handover	5,000	1,500	**6,500**	**INR 4,08,500**	99.63%
W16	Final Project Closeout	1,000	500	**1,500**	**INR 4,10,000**	100.00%

Figure 2.5: Budget Breakdown: Control Account Donut & Total Budget Composition



Financial visualization showing the Cost Baseline (BAC = INR 4,10,000) distributed across 5 Control Accounts, alongside the Contingency Reserve (INR 41,000) and Management Reserve (INR 34,000) forming the Total Budget of INR 4,85,000.

Figure 2.6: Time-Phased Cumulative Expenditure S-Curve Across 16 Weeks



Time-phased cumulative spend S-Curve showing weekly capital drawdown against the approved Cost Baseline (BAC = INR 4,10,000). The curve follows classic project lifecycle dynamics: slow initial startup spend, accelerated development burn during Weeks 4-9, and asymptotic leveling during testing and handover.

5. Academic References & IEEE Bibliography

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